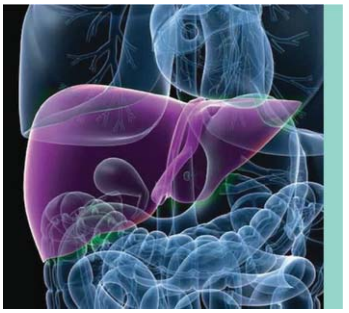


Congestive Hepatopathy: Differentiating Congestion From Fibrosis

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CONGESTIVE HEPATOPATHY

Congestive hepatopathy arises from chronically elevated hepatic venous pressures secondary to right-sided heart failure. Elevated cardiac pressures are transmitted to the central veins of the liver and over time cause presinusoidal dilation, decreased hepatic artery blood flow, and decreased arterial oxygen saturation that leads to bridging fibrosis, cirrhosis, and even hepatocellular carcinoma (HCC).[1-4] An ideal and frequently studied model for congestive liver disease is Fontan-associated liver disease, where patients have liver congestion for decades after receiving a cardiac Fontan operation for single ventricle congenital heart defects (Fig. 1).[5] The Fontan operation provides a dramatic improvement in quality of life and mortality in the early decades of life, but over time all Fontan procedures begin to fail, with a 30-year survival rate of approximately 43%, usually associated with renal and hepatic end-organ dysfunction.[6] Unlike the inflammatory hepatopathies (e.g., viral hepatitis, autoimmune hepatitis, alcoholic and nonalcoholic steatohepatitis), congestive liver disease is a noninflammatory process where early clinical symptoms arise from portal hypertension in the setting of preserved synthetic function.[7,8] A noninflamed, continually congested liver leads to unique serum, radiographic, and histological changes (Table 1), and this potentially reversible congestion is commonly difficult to differentiate from irreversible fibrosis.

SERUM, RADIOGRAPHIC, AND HISTOPATHOLOGICAL FINDINGS

Most serum markers (e.g., aspartate aminotransferase [AST], alanine aminotransferase [ALT], alkaline phosphatase, bilirubin, prothrombin time, albumin, FibroSURE, and hyaluronic acid) have little utility when diagnosing

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; BNP, brain natriuretic peptide; CT, computed tomography; FNH, focal nodular hyperplasia; GGT, gamma-glutamyltransferase; HCC, hepatocellular carcinoma; INR, international normalized ratio; MELD, Model for End-Stage Liver Disease; MELD-XI, Model for End-Stage Liver Disease Without INR; MRE, magnetic resonance elastography; MRI, magnetic resonance imaging.

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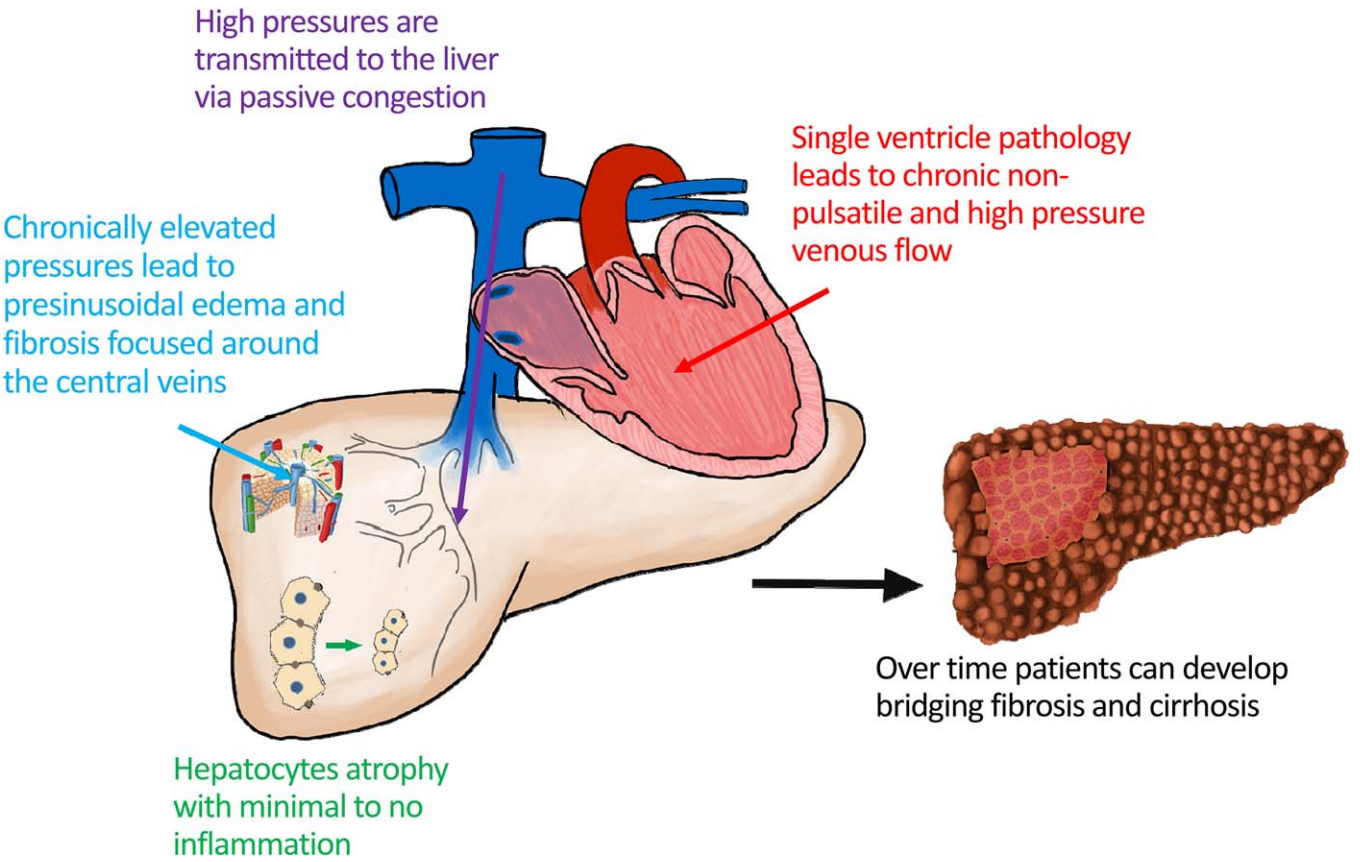


FIG 1 Pathophysiology of Congestive Hepatopathy in Fontan Associated Liver Disease (FALD)[4].

and staging congestive hepatopathy because these values remain close to normal until very end-stage disease and do not correlate with the grade of fibrosis as determined by biopsy.[7-10] Low platelet count and elevated gamma-glutamyltransferase (GGT) levels correlate with the grade of portal hypertension, but not with liver fibrosis, suggesting that they serve as a marker of cardiac function and right-sided pressures rather than liver function.[11,12] Brain natriuretic peptide (BNP) values do not correlate with fibrosis grade on liver biopsy based on one published review.[7] The correlation of BNP values with clinical outcomes in patients with congestive hepatopathy has not been specifically investigated despite some evidence suggesting elevated BNP values correlate with stage of liver dysfunction and mortality in even noncongestive causes of cirrhosis.[13] The Model for End-Stage Liver Disease (MELD) score appears ineffective when estimating severity of liver disease and risk for decompensation in congestive hepatopathy, possibly because the international normalized ratio (INR) increases only in very late-stage disease, and a large cohort of these patients are receiving chronic anticoagulation.[2] The MELD-XI (MELD without INR) was developed to eliminate the variable of anticoagulation in patients with combined cardiac and hepatic dysfunction. The MELD-XI offers more promise for risk stratification in this patient population as multiple studies demonstrate that MELD-XI predicts important clinical outcomes including liver decompensation and death.[14-18] The MELD-XI may also be useful in determining candidacy for heart transplantation or combined heart and liver transplantation, although no specific validation studies for this use currently exist.

Characteristic findings of congestive hepatopathy on imaging include hepatomegaly, dilated venous structures, nodular appearance of the liver, and frequently hyperenhancing nodules. Ultrasound usually demonstrates hepatomegaly and dilation of the inferior vena cava and hepatic veins caused by congestion with increased portal pressure measurements.[3] Computed tomography (CT) and magnetic resonance imaging (MRI) typically show

TABLE 1. SERUM, RADIOGRAPHIC, AND HISTOPATHOLOGICAL CHARACTERISTICS IN CONGESTIVE HEPATOPATHY

Characteristics	Average Values (Range)[5,6,8] or Typical Findings	Clinical Utility
Serum Tests		
AST	31.3 (6-159)	Limited
ALT	28.6 (6-170)	Limited
Alkaline phosphatase	89.8 (7-467)	Limited
Bilirubin	0.83 (0.1-4.8)	Limited
INR	1.19 (0.91-1.77)*	Limited
Albumin	4.43 (1.5-5.5)	Limited
Platelet count	171 (29-659)	Low values can suggest worsening portal hypertension
GGT	72.2 (5-922)	High values can suggest worsening portal hypertension
FibroSURE	0.44 (0.11-0.85), reference: <0.21	Limited
Hyaluronic acid	26 (7-277), reference: <46	Limited
MELD	N/A†	No evidence of use as a clinical risk estimator in congestive liver disease
MELD-XI	N/A†	Moderate evidence for correlation with fibrosis staging by biopsy and for predicting clinical outcomes
Radiographic Studies[3]		
Ultrasound	Hepatomegaly and dilation of inferior vena cava and hepatic veins	Most useful in establishing pressures and flow throughout the portal system
CT and MRI	Hepatomegaly, nodular appearance of liver, frequent arterial nodules usually consistent with regenerative nodules	Useful in identifying lesions that may require biopsy Caution should be used when interpreting nodular appearance as end-stage fibrosis or enhancing lesions as FNH versus HCC
Liver stiffness testing	Uniformly elevated	Limited evidence to suggest any radiographic modality can distinguish congestion versus fibrosis
Histopathological Studies[2]		
Isolated liver biopsy	Presinusoidal edema, pericellular fibrosis focused around the central vein, hepatocyte atrophy without inflammation, later stages with bridging fibrosis and regenerative nodules	Possibly limited because of no accepted fibrosis staging system and heterogeneity of fibrosis
Explanted livers	Significant heterogeneity in fibrosis versus normal parenchyma throughout the liver	Not applicable

*Only the average value from Melero-Ferrer et al.[5] as INR was not reported in other references.
†Was not reported.

hepatomegaly and a nodular appearance of the liver with a fairly high prevalence rate (approximately 30%) of hyperenhancing hepatic nodules that may mimic focal nodular hyperplasia (FNH), but can also be difficult at times to discern from HCC.[3,19,20] A nodular contour of the liver in congestive hepatopathy may not always represent end-stage fibrosis and cirrhosis. Supporting evidence for this theory is that a majority of patients with chronic liver congestion have nodular-appearing livers on imaging, but many of these patients who undergo heart transplantation have no lasting signs of liver dysfunction posttransplant.[2] In addition, there is ongoing debate regarding the management of arterially enhancing hepatic nodules, because it remains unclear whether

the standard radiographic guidelines for distinguishing FNH from HCC apply in congestive liver disease.[21] Therefore, guided biopsies of suspicious lesions may still play a role in confirming the diagnosis of HCC in patients with congestive hepatopathy, at least until further studies confirm or refute that radiographic diagnostic criteria are sufficient in these patients. Liver stiffness assessments using transient elastography, acoustic radiation force impulse elastography, or magnetic resonance elastography (MRE) all uniformly demonstrate elevated values in congestive hepatopathy because of increased blood volumes within the liver and presinusoidal edema.[22-26] These tests appear to be of limited utility to distinguish congestion from fibrosis, although one small study demonstrated moderate correlation of MRE with fibrosis.[23]

Histopathological findings of congestive liver disease include presinusoidal edema, pericellular fibrosis centered around the central veins, and in later stage disease central-to-central and central-to-portal bridging fibrosis with regenerative nodules. Minimal inflammation is present microscopically because hepatocyte death seems to occur through atrophy and apoptosis.[2] No clearly defined staging system exists for fibrosis in congestive hepatopathy because of the atypical pattern of fibrosis development and the heterogeneity of fibrosis deposition throughout the liver. In one series, fibrosis grade on pre–heart transplant liver biopsies had no correlation with postoperative outcomes or hepatic function, bringing into question the utility of biopsy in risk-stratifying patients with congestive hepatopathy for liver-related outcomes.[2]

DIFFERENTIATING CONGESTION FROM FIBROSIS

The biggest challenge in the management of patients with congestive liver disease is differentiating reversible liver congestion from irreversible fibrosis. Currently, there are few validated serum or radiographic tests to reproducibly predict the stage of fibrosis as determined by biopsy, and even isolated liver biopsies may not provide an accurate picture of the total burden of fibrosis in the liver.[2] The development of novel biomarkers and techniques for assessing hepatic fibrosis and function in congestive liver disease is necessary to develop evidence-based management guidelines.

SUMMARY

Congestive hepatopathy (chronic passive venous congestion of the liver) arises from chronically elevated hepatic venous pressures secondary to right-sided heart failure and other cardiac defects. Currently, there are no evidence-based guidelines to direct appropriate screening or management of these patients. The MELD-XI score is the only validated serum-based test to predict clinical outcomes in congestive liver disease, and noninvasive liver stiffness tools seem to be of little utility because all patients have elevated values that currently cannot differentiate between congestion and fibrosis. In congestive hepatopathy, fibrosis staging by liver biopsy does not have an accepted standardized scoring system, fibrosis is quite heterogenous, and pre–heart transplant liver biopsy results are not useful in predicting post–heart transplant hepatic outcomes. Thus, isolated liver biopsies may not provide providers with useful information in caring for these patients. Novel biomarkers to estimate hepatic fibrosis and function are required to fill the gaps in knowledge in this complicated disease.

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REFERENCES

1) Ford RM, Book W, Spivey JR. Liver disease related to the heart. Transplant Rev (Orlando) 2015;29:33-37.

2) Louie CY, Pham MX, Daugherty TJ, Kambham N, Higgins JP. The liver in heart failure: a biopsy and explant series of the histopathologic and laboratory findings with a particular focus on pre-cardiac transplant evaluation. Mod Pathol 2015;28:932-943.

3) Wells ML, Fenstad ER, Poterucha JT, Hough DM, Young PM, Araoz PA, et al. Imaging findings of congestive hepatopathy. Radiographics 2016;36:1024-1037.

4) Hilscher M, Sanchez W. Congestive hepatopathy. Clin Liver Dis 2016;8:68-71.

5) Melero-Ferrer JL, Osa-Sáez A, Buendía-Fuentes F, Ballesta-Cuñat A, Flors L, Rodríguez-Serrano M, et al. Fontan circulation in adult patients: acoustic radiation force impulse elastography as a useful tool for liver assessment. World J Pediatr Congenit Heart Surg 2014;5:365-371.

6) Pundi KN, Johnson JN, Dearani JA, Pundi KN, Li Z, Hinck CA, et al. 40-Year follow-up after the fontan operation: long-term outcomes of 1,052 patients. J Am Coll Cardiol 2015;66:1700-1710.

7) Surrey LF, Russo P, Rychik J, Goldberg DJ, Dodds K, O'Byrne ML, et al. Prevalence and characterization of fibrosis in surveillance liver

biopsies of patients with Fontan circulation. Hum Pathol 2016;57:106-115.

8) Wu FM, Kogon B, Earing MG, Aboulhosn JA, Broberg CS, John AS, et al; Alliance for Adult Research in Congenital Cardiology (AARCC) Investigators. Liver health in adults with Fontan circulation: a multicenter cross-sectional study. J Thorac Cardiovasc Surg 2017;153:656-664.

9) Goldberg DJ, Surrey LF, Glatz AC, Dodds K, O'Byrne ML, Lin HC, et al. Hepatic fibrosis is universal following fontan operation, and severity is associated with time from surgery: a liver biopsy and hemodynamic study. J Am Heart Assoc 2017;6:e004809.

10) Wu FM, Earing MG, Aboulhosn JA, Johncilla ME, Singh MN, Odze RD, et al. Predictive value of biomarkers of hepatic fibrosis in adult Fontan patients. J Heart Lung Transplant 2017;36:211-219.

11) Kaulitz R, Haber P, Sturm E, Schafer J, Hofbeck M. Serial evaluation of hepatic function profile after Fontan operation. Herz 2014;39:98-104.

12) Vasconcelos LA, de Almeida EA, Bachur LF. Clinical evaluation and hepatic laboratory assessment in individuals with congestive heart failure. Arq Bras Cardiol 2007;88:590-595.

13) Wiese S, Mortensen C, Gøtze JP, Christensen E, Andersen O, Bendtsen F, Møller S. Cardiac and proinflammatory markers predict prognosis in cirrhosis. Liver Int 2014;34:e19-e30.

14) Assenza GE, Graham DA, Landzberg MJ, Valente AM, Singh MN, Bashir A, et al. MELD-XI score and cardiac mortality or transplantation in patients after Fontan surgery. Heart 2013;99:491-496.

15) Berg CJ, Bauer BS, Hageman A, Aboulhosn JA, Reardon LC. Mortality risk stratification in fontan patients who underwent heart transplantation. Am J Cardiol 2017;119:1675-1679.

16) Deo SV, Al-Kindi SG, Altarabsheh SE, Hang D, Kumar S, Ginwalla MB, et al. Model for end-stage liver disease excluding international normalized ratio (MELD-XI) score predicts heart transplant outcomes: evidence from the registry of the United Network for Organ Sharing. J Heart Lung Transplant 2016;35:222-227.

17) Farr M, Mitchell J, Lippel M, Kato TS, Jin Z, Ippolito P, et al. Combination of liver biopsy with MELD-XI scores for post-transplant outcome prediction in patients with advanced heart failure and suspected liver dysfunction. J Heart Lung Transplant 2015;34:873-882.

18) Grimm JC, Magruder JT, Do N, Spinner JA, Dungan SP, Kilic A, et al. Modified Model for End-Stage Liver Disease eXcluding INR (MELD-XI) score predicts early death after pediatric heart transplantation. Ann Thorac Surg 2016;101:730-735.

19) Bulut OP, Romero R, Mahle WT, McConnell M, Braithwaite K, Shehata BM, et al. Magnetic resonance imaging identifies unsuspected liver abnormalities in patients after the Fontan procedure. J Pediatr 2013;163:201-206.

20) Wallihan DB, Podberesky DJ. Hepatic pathology after Fontan palliation: spectrum of imaging findings. Pediatr Radiol 2013;43:330-338.

21) Oh C, Youn JK, Han JW, Kim GB, Kim HY, Jung SE. Hepatocellular carcinoma after the Fontan procedure in a 16-year-old girl: a case report. Medicine (Baltimore) 2016;95:e4823.

22) Friedrich-Rust M, Koch C, Rentzsch A, Sarrazin C, Schwarz P, Herrmann E, et al. Noninvasive assessment of liver fibrosis in patients with Fontan circulation using transient elastography and biochemical fibrosis markers. J Thorac Cardiovasc Surg 2008;135:560-567.

23) Poterucha JT, Johnson JN, Qureshi MY, O'Leary PW, Kamath PS, Lennon RJ, et al. Magnetic resonance elastography: a novel technique for the detection of hepatic fibrosis and hepatocellular carcinoma after the fontan operation. Mayo Clin Proc 2015;90:882-894.

24) Wu FM, Opotowsky AR, Raza R, Harney S, Ukomadu C, Landzberg MJ, et al. Transient elastography may identify Fontan patients with unfavorable hemodynamics and advanced hepatic fibrosis. Congenit Heart Dis 2014;9:438-447.

25) Yin M, Glaser KJ, Manduca A, Mounajjed T, Malhi H, Simonetto DA, et al. Distinguishing between hepatic inflammation and fibrosis with MR elastography. Radiology 2017;284:694-705.

26) Yoo BW, Choi JY, Eun LY, Park HK, Park YH, Kim SU. Congestive hepatopathy after Fontan operation and related factors assessed by transient elastography. J Thorac Cardiovasc Surg 2014;148:1498-1505.